

AMBALIKA JAISWAL

Los Angeles, CA | +1 (213) 725-1520 | jaiswalambalika@gmail.com | [LinkedIn](#)

TECHNICAL SKILLS

Languages & Frameworks: Python, Go, JavaScript, TypeScript, SQL, C++, React, React Native, Flask, HTML/CSS

Web & API Development: RESTful APIs, API Gateway, end-to-end web application development, OOP design patterns

ML / AI & Data: PyTorch, Hugging Face Transformers, LangChain, NLP, Sentiment Analysis, Generative AI, Agentic AI, Prompt Engineering, NumPy, pandas, ETL, EDA, Hadoop, Spark, Claude, Copilot, Amazon Q, Cedric, Kiro, Miro

Cloud & Databases: AWS (Lambda, SQS, DLQ, API Gateway, DynamoDB, CloudFront, CDK, SageMaker, Bedrock), Google Firebase, Redis, MongoDB, MySQL, DB2, SQL Server

DevOps & Tooling: Docker, Linux, CI/CD, Azure Pipelines, Git, Unit Testing, Code Coverage, Splunk, Postman, Xcode

EDUCATION

University of Southern California

M.S. Applied Data Science | GPA: 3.72

Aug 2024 – May 2026

Los Angeles, CA

Relevant coursework: Machine Learning for Data Science, Data Mining, Information Retrieval, Foundations of Data Management, Database systems, Data visualization

SRM Institute of Science and Technology

B.Tech Computer Science & Engineering | GPA: 3.83

Jun 2018 – May 2022

Chennai, India

EXPERIENCE

Amazon

Software Development Engineer Intern

May 2025 – Aug 2025

Culver City, CA

- Built and shipped an **AI-powered conversation summarizer** that transforms how users discover and engage with video content, directly impacting 45% of total platform views and driving measurable lift in user engagement across the platform.
- Developed test strategy and ensured production quality via **unit testing, code reviews, and CI/CD integration**; drove code coverage metrics using Amazon Q and Cedric to generate test cases and accelerate reviews.
- Designed a **fault-tolerant distributed serverless pipeline** using asynchronous design patterns (AWS CDK, Lambda, SQS + DLQ, DynamoDB), achieving 95% fault tolerance and zero data loss under load.
- Collaborated with an engineering team to build **prompt-driven LLM pipelines** that mine historical chat threads for sentiment, top product mentions, and key feedback, aiming to cut content discovery time by 40% via seamless RESTful API unification.
- Integrated AI-generated summaries into a **multi-tier scalable application (Amazon Native video player)** via API Gateway and React Native; validated end-to-end on iOS using Xcode, targeting a projected 15% increase in watch time.

Optum (UnitedHealth Group)

Associate Software Engineer II

May 2022 – Jun 2024

Bangalore, India

- Led a 3-person cross-functional Agile team to **refactor automation logic and streamline deployments** via Azure Pipelines and Git, cutting database load and improving pipeline speed by 30%.
- Delivered two **end-to-end Python and SQL ETL automation pipelines** that recovered unprocessable insurance provider records, lifting inventory data quality and coverage from 60% to 85%.
- Engineered a **DB2 + RESTful API solution** to detect and reprocess falsely adjusted claims, achieving 90% successful daily inventory processing and substantial cost avoidance.
- Resolved REST API integration issues in Python, improving data matching accuracy by 25%; restored CI/CD deployments and monitored pipeline telemetry via **Splunk dashboards**.

PROJECTS

Dietary Maps AI | Next.js 14, React 18, TypeScript, Tailwind CSS, Zustand, OpenStreetMap, Kiro, Miro (**Hackathon Winner**)

- Built a **multi-agent AI system** using 8 specialized agents to help users discover restaurants matching complex dietary needs (halal, vegan, gluten-free, keto) via natural language queries.
- Engineered a **transparency confidence scoring pipeline** that rates each restaurant result based on verified OSM dietary tags, evidence strength, data recency, and community signals, eliminating guesswork for users with strict dietary restrictions.

ReviewLens | Python, Hugging Face Transformers, PyTorch

- Engineered a conversational AI system for **multilingual sentiment analysis** that auto-detects review language, classifies intent, and delivers natural language explanations in the user's native language with low-latency, autonomous response generation.

ZeroTrace | Go, TypeScript, React, net/http, goroutines, REST APIs

- Architected a **full-stack zero-trust secret-sharing application** with a TypeScript/React frontend and a Go backend; enforced TTL-based credential expiration, concurrent goroutines, thread-safe in-memory storage, and least-privilege IAM access with automatic secret destruction.

PUBLICATIONS

"Heart Disease Prediction and Detection System" by Jaiswal, A., Purwar, G., C. Pretty. IJSIET, September 2022, Chennai.